Cell: Structure and Functions

A detailed description of the form and appearance of living organisms only brought out their diversity. It is the cell theory that emphasised the unity underlying it, the cellular organisation of all life forms.

The mystery cell theory created

- a description of cell structure and cell growth by division is given in the chapters of this unit
- cell theory also created a sense of mystery around living phenomena, that is around physiological and behavioural processes
- the mystery was the requirement of integrity of cellular organisation for those phenomena to be observed

The physico-chemical approach

- physiological and behavioural processes can be studied by a physico-chemical approach, using cell-free systems
- this describes the processes in molecular terms, and is established by analysis of living tissues for elements and compounds
- it tells us what organic compounds are present, and what they do inside a cell
- and in what way they carry out gross physiological processes such as digestion, excretion, memory, defense and recognition

Reductionist biology

- in other words, what is the molecular basis of all physiological processes, which also explains abnormal processes in any diseased condition
- this physico-chemical approach to study and understand living organisms is called Reductionist Biology
- the concepts and techniques of physics and chemistry are applied to biology

In this unit

Ch 8 Cell: The Unit of Life

Ch 9 Biomolecules

Ch 10 Cell Cycle and Cell Division

G.N. Ramachandran, 1922 to 2001

- an outstanding figure in the field of protein structure, and founder of the Madras school of conformational analysis of biopolymers
- born on October 8, 1922 in a small town not far from Cochin on the southwestern coast of India
- his father was a professor of mathematics at a local college and had considerable influence in shaping his interest in mathematics
- he passed away at the age of 78, on April 7, 2001

His training

- graduated in 1942 as the top-ranking student in the B.Sc. Honors Physics course of the University of Madras
- received a Ph.D. from Cambridge University in 1949
- at Cambridge he met Linus Pauling and was deeply influenced by his publications on models of the alpha-helix and beta-sheet structures, which directed his attention to solving the structure of collagen

What he is remembered for

- his discovery of the triple helical structure of collagen, published in Nature in 1954
- his analysis of the allowed conformations of proteins through the use of the Ramachandran plot
- both rank among the most outstanding contributions in structural biology

What makes a thing living

- looking around, you see both living and non-living things
- what is it that makes an organism living, or that an inanimate thing does not have which a living thing has
- the answer is the presence of the basic unit of life, the cell, in all living organisms
- all organisms are composed of cells, some of a single cell and so unicellular, others, like us, of many cells and so multicellular

8.1 What is a cell?

The fundamental unit

- unicellular organisms are capable of independent existence and of performing the essential functions of life
- anything less than a complete structure of a cell does not ensure independent living
- hence the cell is the fundamental structural and functional unit of all living organisms

Who saw it first

- Antonie Von Leeuwenhoek first saw and described a live cell
- Robert Brown later discovered the nucleus
- the invention of the microscope and its improvement, leading to the electron microscope, revealed all the structural details of the cell

8.2 Cell theory

Schleiden and Schwann

- in 1838 Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant

Schwann, and the final shape of the theory

- Theodore Schwann (1839), a German zoologist, studied different types of animal cells and reported that cells had a thin outer layer, today known as the plasma membrane
- from his studies on plant tissues he concluded that the presence of a cell wall is a unique character of the plant cells, and proposed that the bodies of animals and plants are composed of cells and products of cells
- Schleiden and Schwann together formulated the cell theory, which however did not explain how new cells were formed
- Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells, Omnis cellula-e cellula, giving the theory its final shape

Cell theory as understood today

- all living organisms are composed of cells and products of cells
- all cells arise from pre-existing cells

8.3 An overview of the cell

Eukaryotic and prokaryotic

- the onion cell, a typical plant cell, has a distinct cell wall as its outer boundary and just within it the cell membrane, while human cheek cells have an outer membrane as the delimiting structure
- inside each cell is a dense membrane bound structure, the nucleus, containing the chromosomes which in turn contain the genetic material, DNA
- cells with membrane bound nuclei are eukaryotic, those lacking them prokaryotic
- in both, a semi-fluid matrix called cytoplasm occupies the volume of the cell, and is the main arena of cellular activities where various chemical reactions keep the cell in the living state

Organelles

- besides the nucleus, eukaryotic cells have membrane bound organelles such as the endoplasmic reticulum, golgi complex, lysosomes, mitochondria, microbodies and vacuoles, all of which prokaryotic cells lack
- ribosomes are non-membrane bound organelles found in all cells, both eukaryotic and prokaryotic, occurring in the cytoplasm, within chloroplasts and mitochondria, and on rough ER
- animal cells contain another non-membrane bound organelle, the centrosome, which helps in cell division

Columnar epithelial cells, long and narrow

White blood cells, amoeboid

Red blood cells, round and biconcave

Nerve cell, branched and long

A tracheid, elongated

Mesophyll cells, round and oval

Figure 8.A Diagram showing different shapes of the cells

Cells differ greatly in size

- mycoplasmas, the smallest cells, are only 0.3 µm in length, while bacteria could be 3 to 5 µm
- the largest isolated single cell is the egg of an ostrich
- among multicellular organisms, human red blood cells are about 7.0 µm in diameter, and nerve cells are some of the longest cells

And in shape

- they may be disc-like, polygonal, columnar, cuboid, thread like, or even irregular
- the shape of the cell may vary with the function it performs

8.4 Prokaryotic cells

Who they are

- prokaryotic cells are represented by bacteria, blue-green algae, mycoplasma and PPLO, the Pleuro Pneumonia Like Organisms
- they are generally smaller and multiply more rapidly than eukaryotic cells, and vary greatly in shape and size
- the four basic shapes of bacteria are bacillus or rod like, coccus or spherical, vibrio or comma shaped, and spirillum or spiral
- the organisation of the prokaryotic cell is fundamentally similar even though prokaryotes show a wide variety of shapes and functions

A typical eukaryotic cell, 10-20 µm

Typical bacteria, 1-2 µm

PPLO, about 0.1 µm

Viruses, 0.02-0.2 µm

Figure 8.B Diagram showing comparison of eukaryotic cell with other organisms

What a prokaryote has

- all prokaryotes have a cell wall surrounding the cell membrane, except in mycoplasma, and the semi-fluid matrix filling the cell is the cytoplasm
- there is no well-defined nucleus, the genetic material being basically naked and not enveloped by a nuclear membrane
- besides the genomic DNA, the single circular chromosome, many bacteria have small circular DNA called plasmids, which confer unique phenotypic characters such as resistance to antibiotics, and are used to monitor bacterial transformation with foreign DNA
- no organelles like those of eukaryotes are found except ribosomes, and prokaryotes have something unique in the form of inclusions

8.5 The cell envelope

Three layers acting as one

- most prokaryotic cells, particularly bacterial cells, have a chemically complex cell envelope
- it consists of a tightly bound three layered structure, the outermost glycocalyx, then the cell wall, then the plasma membrane
- although each layer performs a distinct function, they act together as a single protective unit
- bacteria are classified by their envelopes and their response to the staining procedure developed by Gram, those taking up the stain being Gram positive and the others Gram negative

Each layer

- glycocalyx differs in composition and thickness, being a loose sheath called the slime layer in some and a thick tough capsule in others
- the cell wall determines the shape of the cell and gives strong structural support, preventing the bacterium from bursting or collapsing
- the plasma membrane is selectively permeable and interacts with the outside world, being structurally similar to that of eukaryotes
- the mesosome is a special membranous structure formed by extensions of the plasma membrane into the cell, as vesicles, tubules and lamellae

What mesosomes and flagella do

- mesosomes help in cell wall formation, DNA replication and distribution to daughter cells, and also in respiration and secretion, increasing the surface area of the plasma membrane and its enzymatic content
- in some prokaryotes such as cyanobacteria there are other membranous extensions into the cytoplasm called chromatophores, which contain pigments
- bacterial cells may be motile or non-motile, the motile ones having thin filamentous extensions from the cell wall called flagella, which vary in number and arrangement
- the bacterial flagellum is composed of three parts, filament, hook and basal body, the filament being the longest portion, extending from the cell surface to the outside

Pili and fimbriae

- also surface structures of the bacteria, but they play no role in motility
- the pili are elongated tubular structures made of a special protein
- the fimbriae are small bristle like fibres sprouting out of the cell, known in some bacteria to help attach them to rocks in streams and to host tissues

8.6 Ribosomes and inclusion bodies

Prokaryotic ribosomes

- in prokaryotes ribosomes are associated with the plasma membrane, about 15 nm by 20 nm in size, made of two subunits, 50S and 30S, which together form the 70S prokaryotic ribosome
- ribosomes are the site of protein synthesis
- several ribosomes may attach to a single mRNA and form a chain called a polyribosome or polysome, whose ribosomes translate the mRNA into proteins

Inclusion bodies

- reserve material in prokaryotic cells is stored in the cytoplasm as inclusion bodies
- these are not bound by any membrane system and lie free in the cytoplasm, for example phosphate granules, cyanophycean granules and glycogen granules
- gas vacuoles are found in blue green and purple and green photosynthetic bacteria

8.7 Eukaryotic cells

What sets them apart

- the eukaryotes include all the protists, plants, animals and fungi
- there is extensive compartmentalisation of cytoplasm through membrane bound organelles, and an organised nucleus with a nuclear envelope
- they have a variety of complex locomotory and cytoskeletal structures, and their genetic material is organised into chromosomes
- plant cells possess cell walls, plastids and a large central vacuole, absent in animal cells, while animal cells have centrioles, absent in almost all plant cells

The plant cell

- cell wall and middle lamella outside the plasma membrane
- plasmodesmata crossing the wall between neighbouring cells
- a large vacuole, and chloroplasts in the cytoplasm
- nucleus with nucleolus, within the nuclear envelope
- rough and smooth endoplasmic reticulum, golgi apparatus, lysosome
- mitochondrion, peroxisome and microtubule

The animal cell

- plasma membrane as the outer boundary, with microvilli
- centriole, which the plant cell lacks
- nucleus with nucleolus, within the nuclear envelope
- rough and smooth endoplasmic reticulum, golgi apparatus, lysosome
- mitochondrion, peroxisome and ribosomes
- cytoplasm filling the volume of the cell

Figure 8.C Diagram showing : (a) Plant cell (b) Animal cell

Phospholipid bilayer, polar heads out and hydrophobic tails within

Cholesterol within the bilayer

Peripheral protein, lying on the surface

Integral protein, partially or totally buried

Sugar, on the outer face

Figure 8.D Fluid mosaic model of plasma membrane

8.8 The cell membrane

What it is made of

- the detailed structure was studied only after the advent of the electron microscope in the 1950s, though chemical studies, especially on human red blood cells, had let scientists deduce the possible structure
- the cell membrane is mainly composed of lipids and proteins, the major lipids being phospholipids arranged in a bilayer
- the lipids sit with the polar head towards the outer sides and the hydrophobic tails towards the inner part, so that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment
- in addition to phospholipids the membrane also contains cholesterol

Proteins and carbohydrate

- biochemical investigation revealed that cell membranes also possess protein and carbohydrate
- the ratio of protein to lipid varies considerably between cell types, the human erythrocyte membrane having approximately 52 per cent protein and 40 per cent lipids
- by ease of extraction the membrane proteins are classified as integral and peripheral
- peripheral proteins lie on the surface of the membrane, while integral proteins are partially or totally buried in it

The fluid mosaic model

- an improved model was proposed by Singer and Nicolson (1972), widely accepted as the fluid mosaic model
- the quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer, and this ability to move within the membrane is measured as its fluidity
- the fluid nature also matters for functions such as cell growth, formation of intercellular junctions, secretion, endocytosis and cell division

Transport across the membrane

- one of the most important functions of the plasma membrane is the transport of molecules across it, the membrane being selectively permeable
- many molecules move briefly across without any requirement of energy, which is passive transport, neutral solutes moving by simple diffusion along the concentration gradient
- water may also move from higher to lower concentration, and movement of water by diffusion is osmosis
- polar molecules cannot pass through the nonpolar lipid bilayer and require a carrier protein to facilitate their transport

Active transport

- a few ions or molecules are transported against their concentration gradient, from lower to higher concentration
- such transport is an energy dependent process in which ATP is utilised, and is called active transport, for example the Na+/K+ pump

8.9 The cell wall

A non-living rigid structure

- it forms an outer covering for the plasma membrane of fungi and plants
- it gives shape to the cell, protects it from mechanical damage and infection, helps in cell-to-cell interaction and provides a barrier to undesirable macromolecules
- algae have a cell wall of cellulose, galactans, mannans and minerals such as calcium carbonate, while in other plants it consists of cellulose, hemicellulose, pectins and proteins

Primary wall, secondary wall, middle lamella

- the primary wall of a young plant cell is capable of growth, which gradually diminishes as the cell matures, and the secondary wall forms on the inner side
- the middle lamella is a layer mainly of calcium pectate which holds neighbouring cells together
- the cell wall and middle lamellae may be traversed by plasmodesmata, which connect the cytoplasm of neighbouring cells

Nucleus with nuclear pore

Rough endoplasmic reticulum, bearing ribosomes

Smooth endoplasmic reticulum, without them

Figure 8.E Endoplasmic reticulum

Cisternae, flat disc-shaped sacs stacked parallel

Convex cis or forming face

Concave trans or maturing face

Figure 8.F Golgi apparatus

8.10 The endomembrane system

Which organelles belong

- each membranous organelle is distinct in structure and function, but many are considered together as an endomembrane system because their functions are coordinated
- the system includes the endoplasmic reticulum, golgi complex, lysosomes and vacuoles
- mitochondria, chloroplast and peroxisomes are not part of it, since their functions are not coordinated with those components

8.11 The endoplasmic reticulum

Rough and smooth

- electron microscopy reveals a network or reticulum of tiny tubular structures scattered in the cytoplasm, which divides the intracellular space into luminal and extra luminal compartments
- ER bearing ribosomes on its outer surface is rough endoplasmic reticulum, and in their absence it appears smooth and is smooth endoplasmic reticulum
- RER is frequently seen in cells actively involved in protein synthesis and secretion, and is extensive and continuous with the outer membrane of the nucleus
- SER is the major site for synthesis of lipid, and in animal cells lipid-like steroidal hormones are synthesised there

8.12 The golgi apparatus

Cisternae

- Camillo Golgi (1898) first observed densely stained reticular structures near the nucleus, later named golgi bodies after him
- they consist of many flat, disc-shaped sacs or cisternae of 0.5 µm to 1.0 µm diameter, stacked parallel to each other, in varied number
- the cisternae are concentrically arranged near the nucleus with a distinct convex cis or forming face

The two faces

- the cisternae also have a concave trans or maturing face
- the cis and trans faces are entirely different, but interconnected

What the golgi apparatus does

- it principally packages materials, to be delivered either to intra-cellular targets or secreted outside the cell
- materials to be packaged arrive as vesicles from the ER, fuse with the cis face and move towards the maturing face, which is why the golgi remains in close association with the ER
- a number of proteins synthesised by ribosomes on the ER are modified in the cisternae before release from the trans face
- it is the important site of formation of glycoproteins and glycolipids

8.13 Lysosomes and vacuoles

Lysosomes

- membrane bound vesicular structures formed by the process of packaging in the golgi apparatus
- isolated lysosomal vesicles are very rich in almost all types of hydrolytic enzymes, the hydrolases, that is lipases, proteases and carbohydrases, optimally active at acidic pH
- these enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids

Vacuoles

- the vacuole is the membrane-bound space found in the cytoplasm, containing water, sap, excretory product and other materials not useful for the cell
- it is bound by a single membrane called the tonoplast, and in plant cells vacuoles can occupy up to 90 per cent of the volume of the cell
- in plants the tonoplast moves ions and other materials against concentration gradients into the vacuole, so their concentration there is significantly higher than in the cytoplasm
- in Amoeba the contractile vacuole is important for osmoregulation and excretion, and in many cells, as in protists, food vacuoles form by engulfing food particles

8.14 Mitochondria

Number, shape and size

- unless specifically stained they are not easily visible under the microscope
- the number per cell varies with the physiological activity of the cell, and shape and size vary considerably too
- typically sausage-shaped or cylindrical, with a diameter of 0.2 to 1.0 μm, average 0.5 μm, and length 1.0 to 4.1 μm

Outer membrane, the continuous limiting boundary

Inner membrane, folded into cristae

Inter-membrane space between the two

Matrix, the dense substance filling the inner compartment

Figure 8.G Structure of mitochondrion (Longitudinal section)

A double membrane structure

- the outer and inner membranes divide the lumen distinctly into an outer and an inner compartment, the inner one filled with a dense homogeneous substance called the matrix
- the outer membrane forms the continuous limiting boundary of the organelle, while the inner membrane forms infoldings called the cristae towards the matrix, which increase the surface area
- the two membranes have their own specific enzymes associated with mitochondrial function

What they do

- mitochondria are the sites of aerobic respiration, producing cellular energy in the form of ATP, hence they are called the power houses of the cell
- the matrix possesses a single circular DNA molecule, a few RNA molecules, ribosomes of 70S, and the components required for the synthesis of proteins
- the mitochondria divide by fission

8.15 Plastids

Three kinds

- plastids are found in all plant cells and in euglenoides, and are easily observed under the microscope as they are large; they bear specific pigments that impart specific colours
- chloroplasts contain chlorophyll and carotenoid pigments, responsible for trapping the light energy essential for photosynthesis
- chromoplasts carry fat soluble carotenoid pigments such as carotene and xanthophylls, giving the part of the plant a yellow, orange or red colour
- leucoplasts are colourless plastids of varied shapes and sizes with stored nutrients, the amyloplasts storing carbohydrates as in potato, elaioplasts storing oils and fats

Stroma, the space within the inner membrane

Thylakoids, flattened membranous sacs in the stroma

Grana, thylakoids stacked like piles of coins

Stroma lamellae, tubules connecting thylakoids of different grana

Figure 8.H Sectional view of chloroplast

Larger subunit and smaller subunit together

80S in eukaryotes, of 60S and 40S

70S in prokaryotes, of 50S and 30S

Figure 8.I Ribosome

The chloroplast

- aleuroplasts store proteins; the majority of chloroplasts of green plants are found in the mesophyll cells of the leaves
- they are lens-shaped, oval, spherical, discoid or even ribbon-like, of variable length 5 to 10 μm and width 2 to 4 μm, numbering 1 per cell in the green alga Chlamydomonas and 20 to 40 per cell in the mesophyll
- like mitochondria they are double membrane bound, the inner membrane being relatively less permeable, and the space it limits is called the stroma
- the stroma contains enzymes required for the synthesis of carbohydrates and proteins, small double-stranded circular DNA molecules and ribosomes, the chloroplast ribosomes at 70S being smaller than the cytoplasmic 80S

8.16 Ribosomes

Granular structures

- first observed under the electron microscope as dense particles by George Palade (1953)
- composed of ribonucleic acid and proteins, and not surrounded by any membrane
- eukaryotic ribosomes are 80S and prokaryotic 70S, each with a larger and a smaller subunit, the 80S of 60S and 40S and the 70S of 50S and 30S
- S, the Svedberg unit, stands for the sedimentation coefficient and is indirectly a measure of density and size

Cytoskeleton

- an elaborate network of filamentous proteinaceous structures in the cytoplasm, consisting of microtubules, microfilaments and intermediate filaments
- it is involved in many functions such as mechanical support, motility, and maintenance of the shape of the cell

Plasma membrane covering the whole structure

Peripheral microtubules, nine radially arranged doublets

Central microtubule pair, within a central sheath

Radial spoke joining sheath to each doublet, and interdoublet bridges

Figure 8.K Section of cilia/flagella showing different parts : (a) Electron micrograph (b) Diagrammatic representation of internal structure

8.17 Cilia and flagella

Hair-like outgrowths

- cilia and flagella are hair-like outgrowths of the cell membrane
- cilia are small structures which work like oars, causing movement of either the cell or the surrounding fluid, while flagella are comparatively longer and responsible for cell movement
- prokaryotic bacteria also possess flagella, but these are structurally different from eukaryotic flagella

The 9+2 array

- electron microscopy shows they are covered with plasma membrane, their core, the axoneme, possessing microtubules running parallel to the long axis
- the axoneme usually has nine doublets of radially arranged peripheral microtubules and a pair of centrally located ones, an arrangement referred to as the 9+2 array
- the central tubules are connected by bridges and enclosed by a central sheath, connected to one tubule of each peripheral doublet by a radial spoke, so there are nine radial spokes
- the peripheral doublets are also interconnected by linkers, and both cilium and flagellum emerge from a centriole-like structure called the basal body

8.18 Centrosome and centrioles

Two cylinders at right angles

- the centrosome is an organelle usually containing two cylindrical structures called centrioles, surrounded by amorphous pericentriolar materials
- both centrioles lie perpendicular to each other, each with an organisation like a cartwheel

Nuclear membrane, two parallel membranes with perinuclear space

Nuclear pore, formed by fusion of the two membranes

Nucleoplasm, the nuclear matrix

Nucleolus, a spherical body within it

Figure 8.L Structure of nucleus

How a centriole is built

- made up of nine evenly spaced peripheral fibrils of tubulin protein, each fibril a triplet, and the adjacent triplets also linked
- the central part of the proximal region is also proteinaceous and called the hub, connected with tubules of the peripheral triplets by radial spokes made of protein
- the centrioles form the basal body of cilia or flagella, and the spindle fibres that give rise to the spindle apparatus during cell division in animal cells

8.19 The nucleus

The interphase nucleus

- first described by Robert Brown as early as 1831, and the material stained by basic dyes was later given the name chromatin by Flemming
- the interphase nucleus, that of a cell when it is not dividing, has highly extended and elaborate nucleoprotein fibres called chromatin, a nuclear matrix, and one or more spherical bodies called nucleoli
- the nuclear envelope consists of two parallel membranes with a space between them of 10 to 50 nm, the perinuclear space, forming a barrier between the materials inside the nucleus and those of the cytoplasm
- the outer membrane usually remains continuous with the endoplasmic reticulum and also bears ribosomes on it

Pores, and the nucleolus

- at a number of places the envelope is interrupted by minute pores formed by the fusion of its two membranes, through which RNA and protein molecules move in both directions
- normally there is only one nucleus per cell, though variations in number are frequently observed, and some mature cells even lack a nucleus, such as erythrocytes of many mammals and sieve tube cells of vascular plants
- the nucleoplasm contains nucleolus and chromatin; the nucleoli are spherical structures whose content is continuous with the rest of the nucleoplasm, as they are not membrane bound
- the nucleolus is a site for active ribosomal RNA synthesis, and larger and more numerous nucleoli are present in cells actively carrying out protein synthesis

Centromere, the primary constriction

Kinetochore, the disc shaped structure on either side of it

Figure 8.M Chromosome with kinetochore

Metacentric, middle centromere, two equal arms

Sub-metacentric, one shorter and one longer arm

Acrocentric, one extremely short and one very long arm

Telocentric, a terminal centromere

Figure 8.N Types of chromosomes based on the position of centromere

8.20 Chromosomes

Chromatin into chromosomes

- the interphase nucleus has a loose and indistinct network of nucleoprotein fibres called chromatin, but during different stages of cell division cells show structured chromosomes in place of the nucleus
- chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA
- a single human cell has approximately two metre long thread of DNA distributed among its forty six, that is twenty three pairs of, chromosomes

Centromere and kinetochore

- every chromosome, visible only in dividing cells, essentially has a primary constriction or centromere, on the sides of which disc shaped structures called kinetochores are present
- the centromere holds the two chromatids of a chromosome
- based on the position of the centromere, chromosomes can be classified into four types

The four types

- metacentric, with a middle centromere forming two equal arms
- sub-metacentric, with the centromere slightly away from the middle, giving one shorter and one longer arm
- acrocentric, with the centromere close to the end, giving one extremely short and one very long arm
- telocentric, with a terminal centromere

Secondary constrictions

- sometimes a few chromosomes have non-staining secondary constrictions at a constant location
- this gives the appearance of a small fragment called the satellite

Microbodies

- many membrane bound minute vesicles called microbodies, containing various enzymes, are present in both plant and animal cells

8.21 Summary

Cells and their parts

- all organisms are made of cells or aggregates of cells, which vary in shape, size and functions; by the presence or absence of a membrane bound nucleus and other organelles, cells and hence organisms are named eukaryotic or prokaryotic
- a typical eukaryotic cell consists of a cell membrane, nucleus and cytoplasm, with plant cells carrying a cell wall outside the membrane
- the plasma membrane is selectively permeable and facilitates transport of several molecules, and the endomembrane system includes ER, golgi complex, lysosomes and vacuoles
- centrosome and centriole form the basal body of cilia and flagella that facilitate locomotion, and in animal cells centrioles also form the spindle apparatus during cell division; the nucleus contains nucleoli and chromatin network, controls the activities of organelles and plays a major role in heredity

The organelles, in brief

- endoplasmic reticulum contains tubules or cisternae, rough and smooth, and helps in transport of substances and synthesis of proteins, lipoproteins and glycogen
- the golgi body is composed of flattened sacs in which cell secretions are packed and transported; lysosomes are single membrane structures containing enzymes for digestion of all types of macromolecules; ribosomes occur freely in the cytoplasm or on the ER and are involved in protein synthesis
- mitochondria help in oxidative phosphorylation and generation of adenosine triphosphate, bound by a double membrane, the outer smooth and the inner folded into cristae
- plastids are pigment containing organelles found in plant cells only, the grana being the site of light reactions and the stroma of dark reactions; green plastids are chloroplasts containing chlorophyll, other coloured ones chromoplasts with carotene and xanthophyll

8.22 Exercises

From this chapter

1. Which of the following is not correct? (a) Robert Brown discovered the cell. (b) Schleiden and Schwann formulated the cell theory. (c) Virchow explained that cells are formed from pre-existing cells. (d) A unicellular organism carries out its life activities within a single cell.
2. New cells generate from (a) bacterial fermentation (b) regeneration of old cells (c) pre-existing cells (d) abiotic materials
3. Match these against one another: (a) Cristae (b) Cisternae (c) Thylakoids, and (i) flat membranous sacs in stroma (ii) infoldings in mitochondria (iii) disc-shaped sacs in golgi apparatus
4. Which of the following is correct? (a) Cells of all living organisms have a nucleus. (b) Both animal and plant cells have a well defined cell wall. (c) In prokaryotes, there are no membrane bound organelles. (d) Cells are formed de novo from abiotic materials.
5. What is a mesosome in a prokaryotic cell? Mention the functions that it performs.
6. How do neutral solutes move across the plasma membrane? Can the polar molecules also move across it in the same way? If not, then how are these transported across the membrane?
7. Name two cell organelles that are double membrane bound. What are the characteristics of these two organelles? State their functions and draw labelled diagrams of both.
8. What are the characteristics of prokaryotic cells?
9. Multicellular organisms have division of labour. Explain.
10. Cell is the basic unit of life. Discuss in brief.
11. What are nuclear pores? State their function.
12. Both lysosomes and vacuoles are endomembrane structures, yet they differ in terms of their functions. Comment.
13. Describe the structure of the following with the help of labelled diagrams. (i) Nucleus (ii) Centrosome
14. What is a centromere? How does the position of centromere form the basis of classification of chromosomes? Support your answer with a diagram showing the position of centromere on different types of chromosomes.